

Literature Review Guide — ACVRP-CO₂

This document provides a structured literature review for the project "**Quantifying the Asymmetry Penalty in Carbon-Aware Last-Mile Routing under Real Asymmetric Road Networks**".

Its purpose is twofold:

1. **Give your student a starting point** — every paper here has been verified to exist, with the citation, the year, and a one-line justification of why it matters for *this specific* paper.
2. **Show them how to organise a literature review** by theme rather than chronology — this is how IJHSR-grade papers structure their Related Work section.

The reading is divided into **five themes**. Each theme has 3–6 key references. The student does not need to read all of them in full — for several, the abstract + intro is enough. The asterisks indicate priority:

- *** Must read in full (foundational or directly cited in our methodology)
 - ** Skim the abstract + intro + figures
 - * Aware of, can cite without deep reading
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Theme 1: Classical Capacitated Vehicle Routing Problem (CVRP)

This theme covers what the CVRP is, how it has been formulated as a mixed-integer program, and where the benchmark instances come from. The student needs to understand this clearly before they can claim that *asymmetric* CVRP is different.

*** Dantzig & Ramser (1959). The Truck Dispatching Problem.

Management Science, 6(1), 80–91. The paper that founded the field. Required citation for any VRP paper. A short note that introduced the truck dispatching problem and showed how it could be formulated as a linear-programming relaxation of an integer program.

*** Uchoa et al. (2017). New Benchmark Instances for the Capacitated Vehicle Routing Problem.

European Journal of Operational Research, 257(3), 845–858. Defines CVRPLIB, the standard benchmark suite that almost every VRP paper uses for comparison. Our paper will cite this when explaining that we deliberately *do not* use these synthetic Euclidean instances because they assume symmetric distances.

** Wouda, Lan & Kool (2024). PyVRP: A High-Performance VRP Solver Package.

INFORMS Journal on Computing, 36(4), 943–955. The current state-of-the-art open-source classical VRP solver (implements Hybrid Genetic Search; won the 2021 DIMACS challenge and the 2022 EURO-NeurIPS competition). Useful as a high-end classical baseline. We use OR-Tools instead because PyVRP currently does not natively support arbitrary asymmetric matrices, but PyVRP is worth naming in Related Work.

*** Christofides, Mingozzi & Toth (1979). The Vehicle Routing Problem. In *Combinatorial Optimization*, Wiley, 315–338.**

The original benchmark set ("CMT" instances). Mentioned briefly as the predecessor to CVRPLIB.

Theme 2: Asymmetric Vehicle Routing (the closest prior work)

This is the most directly relevant theme. The student should be able to articulate exactly what we contribute on top of these papers.

***** Lee & Kim (2021). A Proposal and Analysis of New Realistic Sets of Benchmark Instances for Vehicle Routing Problems with Asymmetric Costs.**

Applied Sciences, 11(11), 4790. Currently the most direct precedent for our work. They built ACVRP benchmark instances using a Korean map API (T map). Important distinction we should highlight: their benchmarks are *synthetic* realisations of a few cities, while we evaluate on OSM-derived instances for two specific cities (Macau, Hong Kong) and quantify the *asymmetry penalty as a function of solver class*. Cite this explicitly when justifying our methodology.

***** Melo, Mota, Andrade & Araújo (2022). The Impact of One-Way Streets on the Asymmetry of the Shortest Commuting Routes.**

Physical Review Research, 4(2), 023053. arXiv:2111.07434. Empirically measures route asymmetry on OSM data for ten cities. Defines a log-ratio metric for asymmetry. Our "asymmetry index" is a related but VRP-specific metric, so we should cite this as the closest precedent and explain the difference (they study individual O-D pairs; we measure how this propagates through the optimal *tour*).

**** Erbao Cao et al. (2020). A New Model for the Asymmetric Vehicle Routing Problem with Simultaneous Pickup and Deliveries.**

Operations Research Letters, 48(2), 161–164. States the importance of asymmetric VRPs in urban contexts with one-way streets or sloped terrain — useful one-line citation for motivation.

*** Coordinated Truck-and-Drone Asymmetric Arc Routing (2022). PMC9415381 / Mathematical Problems in Engineering.**

An example of arc-routing extension that explicitly accounts for one-way streets. Cite once when listing "other asymmetric-routing variants" in Related Work.

Theme 3: Pollution-Routing & Green VRP (the emissions side)

Justifies why CO₂ matters as an objective and how it has been modelled in prior work. Establishes that the linear fuel/emissions model we use is well-grounded.

*** Bektaş & Laporte (2011). The Pollution-Routing Problem.

Transportation Research Part B: Methodological, 45(8), 1232–1250. **The single most important citation for our emissions formulation.** Introduces the Pollution-Routing Problem (PRP) and the load-and-speed- dependent fuel model. Our simplified linear model (distance + time + load-distance) is a tractable specialisation of theirs. The paper also reports up to 7% CO₂ savings achievable by re-routing — this is a natural reference point to put our 4–42% asymmetry penalty against.

*** Demir, Bektaş & Laporte (2014). The Bi-Objective Pollution-Routing Problem.

European Journal of Operational Research, 232(3), 464–478. Extends PRP to multi-objective (fuel + driving time). Our paper's multi-objective framing (distance + CO₂) is methodologically aligned with this work. Cite when introducing the scalarisation $\alpha_1 \cdot \text{distance} + \alpha_2 \cdot \text{CO}_2$.

** Asghari & Mirzapour Al-e-Hashem (2024). Survey of Green Vehicle Routing Problem: Past and Future Trends.

Operations Research Forum / RG review, 2024 update covering 75 PRP papers from 2011 to 2024. Comprehensive review of the green VRP literature. The student should use it as a map to find any sub-topic in the field. **Single best secondary source** for the Related Work paragraph on green VRP.

** Lin, Choy, Ho, Chung & Lam (2014). Survey of Green Vehicle Routing Problem: Past and Future Trends.

Expert Systems with Applications, 41(4), 1118–1138. The other major review (earlier, pre-2014). Useful for citing the historical evolution of Green VRP.

* Demir, Bektaş & Laporte (2012). An Adaptive Large Neighborhood Search Heuristic for the Pollution-Routing Problem.

European Journal of Operational Research, 223(2), 346–359. The follow-up algorithmic paper showing ALNS for PRP. Cite when discussing classical heuristics for emission-aware routing.

Theme 4: Neural Combinatorial Optimization (NCO) for routing

The four foundational papers behind our MatNet-CVRP architecture. Read these in chronological order to see the evolution from RNN+supervised learning to attention+RL.

*** Vinyals, Fortunato & Jaitly (2015). Pointer Networks.

Advances in Neural Information Processing Systems (NIPS) 28. arXiv:1506.03134. First neural architecture able to output a permutation of variable-length input (essential for routing). Sets the foundation for everything that follows. Solved planar TSP up to $n=50$ via supervised learning.

*** Bello, Pham, Le, Norouzi & Bengio (2017). Neural Combinatorial Optimization with Reinforcement Learning.

ICLR Workshop. arXiv:1611.09940. Replaces supervised learning (Vinyals 2015) with REINFORCE, enabling training without optimal labels — critical because optimal CVRP solutions are expensive to obtain. Establishes the basic NCO recipe: encoder + autoregressive pointer decoder + RL objective.

*** Kool, van Hoof & Welling (2019). Attention, Learn to Solve Routing Problems!

ICLR 2019. arXiv:1803.08475. Code: github.com/wouterkool/attention-learn-to-route. **The most important architectural precedent for our Vanilla-AM baseline.** Replaces RNNs with multi-head self-attention, introduces the greedy-rollout REINFORCE baseline. Achieves near-optimal solutions for TSP/CVRP up to $n=100$. Our Vanilla-AM is an exact re-implementation of this architecture (without edge features). Cite explicitly whenever we say "AM" or "the Attention Model".

*** Kwon, Choo, Kim, Yoon, Gwon & Min (2020). POMO: Policy Optimization with Multiple Optima for Reinforcement Learning.

NeurIPS 2020. arXiv:2010.16011. Code: github.com/yd-kwon/POMO. The training recipe we use. Replaces the greedy rollout baseline of Kool 2019 with the multi-start mean baseline (launch K rollouts from K different first customers; the baseline for each is the mean of the others). Significantly faster training and better local-minimum escape. Reduces TSP100 optimality gap to 0.14%. **Note we deliberately do NOT use the rotation/reflection augmentation also introduced in this paper** — those operations corrupt the asymmetric distance matrix. Make this explicit in our methodology section.

*** Kwon, Choo, Yoon, Park, Park & Gwon (2021). Matrix Encoding Networks for Neural Combinatorial Optimization.

NeurIPS 2021. arXiv:2106.11113. Code: github.com/yd-kwon/MatNet. **The most direct architectural precedent for our work.** Introduces the bidirectional attention encoder for matrix-valued inputs (such as asymmetric distance matrices). Applied to asymmetric TSP and flexible flow-shop scheduling. Our MatNet-CVRP extends this idea by: (i) integrating it with capacity constraints (CVRP, not just TSP), (ii) attending to multi-channel edge features (distance, time, fuel, CO_2 — not just a scalar entry), and (iii) replacing the random one-hot positional

encoding with geographic coordinates from OSM. State all three contributions explicitly in our Related Work paragraph on MatNet.

**** Berto et al. (2024). RL4CO: an Extensive Reinforcement Learning for Combinatorial Optimization Benchmark.**

KDD 2025. arXiv:2306.17100. Code: github.com/ai4co/rl4co. The current unified library for NCO research, with implementations of AM, POMO, MatNet, and 20+ other models. Worth citing as the standard reference codebase for the field. Useful for the student to know about even if we built our own implementation.

Theme 5: OSM-Based Routing & Real Road Networks

Documents the tools and conventions for working with real road network data, which is what distinguishes our paper from prior NCO routing work.

***** Boeing (2017). OSMnx: A Python Package to Work with Graph-Theoretic OpenStreetMap Street Networks.**

Journal of Open Source Software, 2(12), 215. The original OSMnx paper. Required citation when describing how we obtain the road network for Macau and Hong Kong. Always cite the JOSS 2017 version (DOI: 10.21105/joss.00215) as it's the canonical peer-reviewed citation.

**** Boeing (2025). Modeling and Analyzing Urban Networks and Amenities with OSMnx.**

Geographical Analysis. DOI: 10.1111/gean.70009. The 2025 update covering the current OSMnx 2.0 capabilities. Cite alongside the 2017 paper if reviewers want the latest reference.

***** Melo et al. (2022). The Impact of One-Way Streets on the Asymmetry of the Shortest Commuting Routes.**

Already listed in Theme 2; deserves to be cited here too because they demonstrated empirically that OSM-derived road graphs *do* exhibit measurable directional asymmetry across major world cities.

How to Use This Guide

Suggested reading order (one focused week of reading)

Day	Theme & Papers
Mon	Theme 1 in full (Dantzig 1959, Uchoa 2017) + skim PyVRP
Tue	Theme 3 in full (Bektaş 2011 most important) + Demir 2014

Day	Theme & Papers
Wed	Theme 4 part 1: Vinyals 2015 + Bello 2017
Thu	Theme 4 part 2: Kool 2019 + POMO 2020
Fri	Theme 4 part 3: MatNet 2021 (re-read more carefully — it directly informs our architecture)
Sat	Theme 2: Lee 2021 + Melo 2022 (asymmetric VRP precedents)
Sun	Theme 5: OSMnx 2017; review and start drafting Related Work

Drafting the "Related Work" section (Section 2 of the paper)

Use this skeleton, with one paragraph per theme:

2.1 Classical CVRP and benchmark conventions. [Use Theme 1 citations to explain the standard formulation, mention that almost all benchmarks assume symmetric 2D Euclidean distances.]

2.2 Asymmetric VRP variants. [Theme 2. State that asymmetric CVRP has received less attention than symmetric. Lee 2021 created realistic ACVRP benchmarks but did not measure the cost of ignoring asymmetry.]

2.3 Carbon-aware vehicle routing. [Theme 3. The Pollution- Routing Problem (Bektaş & Laporte 2011) introduces emissions as an objective. Most of the green-VRP literature uses symmetric Euclidean distances; combining green objectives with realistic asymmetric road topology is rare.]

2.4 Neural combinatorial optimisation for routing. [Theme 4. Chronological narrative from Vinyals 2015 to MatNet 2021. End with the observation that no existing NCO architecture handles both capacity constraints AND asymmetric matrices.]

2.5 Position of this work. [One-paragraph synthesis: we combine the asymmetric matrix encoding of MatNet with the capacity-aware decoder of POMO/AM, train under a CO₂-augmented reward inspired by Bektaş & Laporte, and evaluate on OSM road networks of Macau and Hong Kong. To our knowledge, no prior work addresses this exact intersection.]

BibTeX-friendly citation list

For convenience, this is the full bibliography in @article format — the student can paste this into their reference manager:


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@article{dantzig1959,
  author = {Dantzig, George B. and Ramser, John H.},
  title = {The Truck Dispatching Problem},
  journal= {Management Science},
  volume = {6}, number = {1}, pages = {80--91}, year = {1959}
}

@article{uchoa2017cvrplib,
  author = {Uchoa, E. and Pecin, D. and Pessoa, A. and Poggi, M. and Vidal,
  title = {New Benchmark Instances for the Capacitated Vehicle Routing Prob
  journal= {European Journal of Operational Research},
  volume = {257}, number = {3}, pages = {845--858}, year = {2017}
}

@article{wouda2024pyvrp,
  author = {Wouda, Niels A. and Lan, Leon and Kool, Wouter},
  title = {{PyVRP}: A High-Performance {VRP} Solver Package},
  journal= {INFORMS Journal on Computing},
  volume = {36}, number = {4}, pages = {943--955}, year = {2024},
  doi = {10.1287/ijoc.2023.0055}
}

@article{lee2021acvrp,
  author = {Lee, Jusang and Kim, Byung-In},
  title = {A Proposal and Analysis of New Realistic Sets of Benchmark Insta
  journal= {Applied Sciences},
  volume = {11}, number = {11}, pages = {4790}, year = {2021}
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@article{melo2022asymmetry,
  author = {Melo, Hygor P. M. and Mota, Diogo P. and Andrade, Jr., Jos'e S.
  title = {Impact of one-way streets on the asymmetry of the shortest commu
  journal= {Physical Review Research},
  volume = {4}, number = {2}, pages = {023053}, year = {2022},
  doi = {10.1103/PhysRevResearch.4.023053}
}

@article{bektas2011prp,
  author = {Bekta\c{s}, Tolga and Laporte, Gilbert},
  title = {The Pollution-Routing Problem},
  journal= {Transportation Research Part B: Methodological},
  volume = {45}, number = {8}, pages = {1232--1250}, year = {2011},
  doi = {10.1016/j.trb.2011.02.004}
}

@article{demir2014biobjective,
  author = {Demir, Emrah and Bekta\c{s}, Tolga and Laporte, Gilbert},
  title = {The Bi-Objective Pollution-Routing Problem},
  journal= {European Journal of Operational Research},
  volume = {232}, number = {3}, pages = {464--478}, year = {2014}
}

@article{demir2012alns,
  author = {Demir, Emrah and Bekta\c{s}, Tolga and Laporte, Gilbert},
  title = {An Adaptive Large Neighborhood Search Heuristic for the Pollutio

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    journal= {European Journal of Operational Research},
    volume = {223}, number = {2}, pages = {346--359}, year = {2012}
}

@inproceedings{vinyals2015ptrnet,
  author = {Vinyals, Oriol and Fortunato, Meire and Jaitly, Navdeep},
  title = {Pointer Networks},
  booktitle = {Advances in Neural Information Processing Systems},
  volume = {28}, year = {2015}
}

@misc{bello2017nco,
  author = {Bello, Irwan and Pham, Hieu and Le, Quoc V. and Norouzi, Mohamma},
  title = {Neural Combinatorial Optimization with Reinforcement Learning},
  year = {2017},
  eprint = {1611.09940},
  archivePrefix = {arXiv}
}

@inproceedings{kool2019attention,
  author = {Kool, Wouter and van Hoof, Herke and Welling, Max},
  title = {Attention, Learn to Solve Routing Problems!},
  booktitle = {International Conference on Learning Representations},
  year = {2019}
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@inproceedings{kwon2020pomo,
  author = {Kwon, Yeong-Dae and Choo, Jinho and Kim, Byoungjip and Yoon, Ilj},
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  booktitle = {Advances in Neural Information Processing Systems},
  volume = {33}, year = {2020}
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@inproceedings{kwon2021matnet,
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  title = {Matrix Encoding Networks for Neural Combinatorial Optimization},
  booktitle = {Advances in Neural Information Processing Systems},
  volume = {34}, year = {2021}
}

@inproceedings{berto2024rl4co,
  author = {Berto, Federico and Hua, Chuanbo and Park, Junyoung and others},
  title = {{RL4CO}: an Extensive Reinforcement Learning for Combinatorial O},
  booktitle = {Proceedings of the 31st ACM SIGKDD Conference on Knowledge Di},
  year = {2025},
  eprint = {2306.17100},
  archivePrefix = {arXiv}
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@article{boeing2017osmnx,
  author = {Boeing, Geoff},
  title = {{OSMnx}: A Python Package to Work with Graph-Theoretic {OpenStre}},
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  volume = {2}, number = {12}, pages = {215}, year = {2017},
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@article{boeing2025osmnx,  
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  journal= {Geographical Analysis},  
  year  = {2025},  
  doi   = {10.1111/gean.70009}  
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@article{williams1992reinforce,  
  author = {Williams, Ronald J.},  
  title  = {Simple Statistical Gradient-Following Algorithms for Connectioni},  
  journal= {Machine Learning},  
  volume = {8}, pages = {229--256}, year = {1992}
```

(williams1992reinforce is the original REINFORCE paper — cite once in the methodology section when introducing the policy gradient.)

A Note on Methodology

When the student writes the Related Work section, the key thing is to **make the gap explicit**. The most powerful single sentence to write is something like:

"While prior work has separately studied (i) asymmetric VRP with classical solvers [Lee 2021], (ii) emissions as a VRP objective [Bektaş & Laporte 2011], and (iii) neural combinatorial optimisation for symmetric Euclidean CVRP [Kool 2019, Kwon 2020], to our knowledge no prior work combines all three: a neural solver that handles capacity constraints AND asymmetric real-road matrices AND a CO₂-aware reward, evaluated on actual OSM-derived road networks of dense Asian cities."

That sentence is the **whole justification for the paper**. Everything else in Related Work is supporting evidence.